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Operations Research

Publication details, including instructions for authors and subscription information:
http://pubsonline.informs.org

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To cite this article:
Hanzhao Wang, Kalyan Talluri, Xiaocheng Li (2025) Technical Note—On Dynamic Pricing with Covariates. Operations Research 73(4):1932-1943. https://doi.org/10.1287/opre.2021.0802

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OPERATIONS RESEARCH
Vol. 73, No. 4, July–August 2025, pp. 1932–1943
ISSN 0030-364X (print), ISSN 1526-5463 (online)

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Crosscutting Areas

Technical Note—On Dynamic Pricing with Covariates

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Received: December 25, 2021
Revised: November 13, 2023; July 24, 2024; November 15, 2024
Accepted: December 22, 2024
Published Online in Articles in Advance: January 31, 2025

Area of Review: Market Analytics and Revenue Management

https://doi.org/10.1287/opre.2021.0802

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Abstract. We consider dynamic pricing with covariates under a generalized linear demand model: A seller can dynamically adjust the price of a product over a horizon of T time periods, and at each time period t, the demand of the product is jointly determined by the price and an observable covariate vector $x_t \in \mathbb{R}^d$ through a generalized linear model with unknown coefficients. Most of the existing literature assumes the covariate vectors x_ts are independently and identically distributed (i.i.d.); the few papers that relax this assumption either sacrifice model generality or yield suboptimal regret bounds. In this paper, we show that Upper Confidence Bound and Thompson sampling-based pricing algorithms can achieve an $O(d\sqrt{T}\log T)$ regret upper bound without assuming any statistical structure on the covariates x_t. Our upper bound on the regret matches the lower bound up to logarithmic factors. We thus show that (i) the i.i.d. assumption is not necessary for obtaining low regret, and (ii) the regret bound can be independent of the (inverse) minimum eigenvalue of the covariance matrix of the x_ts, a quantity present in previous bounds. Moreover, we consider a constrained setting of the dynamic pricing problem where there is a limited and unreplenishable inventory, and we develop theoretical results that relate the best achievable algorithm performance to a variation measure with respect to the temporal distribution shift of the covariates. We also demonstrate the proposed algorithms' performance with numerical experiments.

Supplemental Material: All supplemental materials, including the code, data, and files required to reproduce the results, are available at https://doi.org/10.1287/opre.2021.0802.

Keywords: dynamic pricing • online learning • adversarial covariates • regret analysis

1. Introduction

In this paper, we consider the problem of dynamically pricing a product over time when additional covariate information is available. In the literature, this is also referred to as "feature-based dynamic pricing," "personalized dynamic pricing," or "dynamic pricing with side information." Over time periods $t = 1, \ldots, T$, the seller observes covariates $x_t \in \mathbb{R}^d$ in each period, sets a price, and then observes the realized demand. The covariates generally are a set of features that characterize the pricing environment at both a macro level (market or environment) and a micro level (customer-specific). The seller's objective is to maximize revenue and the demand depends on the price and the covariates through a parametric function. The seller needs to balance the learning of the demand function with the earning of the revenue, the classic exploration-exploitation tradeoff. The performance of a pricing policy or algorithm is measured by the gap between the expected cumulative revenue obtained by the policy and that of an optimal policy which knows the demand function a priori.

At a high level, the existing literature on dynamic pricing with covariates can be grouped into three categories:

1. Allow arbitrary covariates but assume both the knowledge of the price-sensitivity coefficient and the parameters of the distribution of the demand shock. In Online Appendix B, we discuss modeling issues associated with this assumption.

2. Assume independently and identically distributed (i.i.d.) covariates and allow an unknown price coefficient and an unknown parameter of the distribution of the demand shock.

3. Allow arbitrary covariates with an unknown price coefficient and parameters of the error distribution but able to derive only a suboptimal regret bound.

The i.i.d. assumption in the second stream above can be restrictive in many situations in practice, say when there is serial correlation or peer effects in the data (we elaborate further in Online Appendix B).

In this paper, we study a setting where the covariates x_t can be arbitrary, that is, no i.i.d. assumption, and the demand depends on the covariates and the price

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through a generalized linear model (GLM). The main contributions of this paper are as follows:

• We give a simple UCB-based pricing algorithm, which makes no assumptions on how the covariates are generated and yet achieves an optimal order of regret. This algorithm is inspired by the GLM-UCB algorithm for generalized linear reward bandits (Filippi et al. 2010). In that framework, the reward is modeled as a generalized linear function of the action. However, our reward function (revenue) does not follow a generalized linear form in the action (price). Consequently, the analysis of our pricing algorithm has to take into account this misalignment between the reward and the observation (demand).

• We next propose a Thompson sampling-based pricing algorithm, which is more computationally efficient than the UCB-based one while still enjoying an optimal order of regret. By leveraging the special structure of the pricing problem, we improve on two known limitations to Thompson sampling for generalized linear bandits (Agrawal and Devanur 2016, Abeille and Lazaric 2017), namely (1) a suboptimal order of regret with respect to the dimension of the covariates and (2) a requirement of convexity of the reward function with respect to the unknown parameters.

• We extend the previous Thompson sampling dynamic pricing algorithm to a setting with an inventory constraint, where the seller has a limited and unreplenishable inventory at the beginning of the horizon. We show the distributional shift of the covariates adds an unavoidable term on the regret. Therefore, we modify the Thompson sampling algorithm into a dual-based algorithm to balance the inventory consumption by adaptively maintaining a dual variable.

The remainder of the paper is organized as follows. Section 1.1 presents a brief review of the literature. Section 2 gives the formal definition of the problem with assumptions and the performance measure. Section 3 designs a UCB-based pricing algorithm with a discussion on its computation efficiency. Section 4 proposes a Thompson sampling pricing algorithm to alleviate the potential computation issue in the UCB pricing algorithm. We further extend the Thompson sampling pricing algorithm to a setting with an inventory constraint in Section 5. Finally, Section 6 provides empirical simulations, and Section 7 concludes the paper.

1.1. Literature Review

Ban and Keskin (2021) consider the generalized linear demand model that includes both the binary and linear demand models as special cases, extending the covariate-free case in Broder and Rusmevichientong (2012). We work with this generalized linear demand model, but our algorithm and analysis differ from that of Ban and Keskin (2021). Qiang and Bayati (2016) impose a martingale-type condition on the covariates that is similar to the i.i.d. assumption because both in essence ensure the minimum eigenvalue of the sample covariance matrix is bounded away from zero. In Online Appendix B.3, we discuss the technical and practical advantages of removing the i.i.d. assumption.

Ferreira et al. (2018) consider the price-based network revenue management problem where there are finitely many allowable prices and a number of resource constraints. Javanmard and Nazerzadeh (2019) and Ban and Keskin (2021) pose their dynamic pricing problem in a high-dimensional setting, with a sparsity assumption, and perform a variable selection subroutine in their pricing algorithm via ℓ_1 regularization. To this end, we believe the i.i.d. assumption and the dependence of the bounds on the minimum eigenvalue in both papers are necessary to handle the high-dimensional setting in contrast to our low-dimensional one where we can remove those assumptions. Keskin et al. (2022) formulate a stylized framework for electricity pricing, where the covariates can be classified into three types based on their different forms of heterogeneity and thus are non-i.i.d. They develop a spectral clustering method to tailor the featured-based pricing, which is quite different from our algorithms. We summarize the existing works in Table 1. For more on the history and origin of the dynamic pricing problem, especially the covariate-free case, we refer the readers to the review paper Den Boer (2015).

Our work is also related to generalized linear contextual bandits (Filippi et al. 2010, Abeille and Lazaric 2017). We utilize techniques from this literature, whereas our work is differentiated in the following aspects: (1) there is a slight misalignment between the reward (revenue) and observation (demand) in the pricing setting, while in the bandits setting they are generally same; (2) in addition, the reward functions in these works are assumed to be generalized linear functions of actions; in contrast, our reward function (revenue) is not generalized linear in the action (price), only the observed demand is; and (3) we further relax the assumptions and improve the regret bounds of Abeille and Lazaric (2017) by leveraging the special structure of the pricing problem.

2. Problem Setup

Consider the generalized linear demand model

$$D_t = g(x_t^\top \alpha^* + x_t^\top \beta^* \cdot p_t) + \epsilon_t \quad \forall t = 1, 2, \ldots, T, \tag{1}$$

where $\alpha^*, \beta^* \in \mathbb{R}^d$ are true unknown parameters, $g(\cdot)$ is a known link function, x_t is an observable covariate vector (i.e., the seller observes x_t before setting the price p_t), and ϵ_t is an unobservable and idiosyncratic demand shock in period t.

The generalized linear model covers the two mainstream demand models, the binary demand model and the linear demand model, as special cases (see Online

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Table 1. Summary of Existing Results

Paper	Regret bound	Covariates	Demand model	Key assumptions
Qiang and Bayati (2016)	$O(d \log T)$	Mart. Diff.	$x^\top \alpha + bp + \epsilon$	With known incumbent price
Cohen et al. (2020)	$O(d^2 \log T)$	Adver.	$\mathbb{1}\{x^\top \alpha \geq p\}$	—
Cohen et al. (2020)	$\tilde{O}(d^{19/6} T^{\frac{2}{3}})$	Adver.	$\mathbb{1}\{x^\top \alpha + \epsilon \geq p\}$	Sub-Gaussian noise
Mao et al. (2018)	$\tilde{O}(T^{\frac{d}{d+1}})$	Adver.	$\mathbb{1}\{f(x) \geq p\}$	$f(\cdot)$ is Lipschitz
Javanmard (2017)	$O(\sqrt{T})$	Adver.	$\mathbb{1}\{x^\top \alpha_t + \epsilon \geq p\}$	Log-concave noise with changing α_t
Javanmard and Nazerzadeh (2019)	$O(d \log T)$	i.i.d.	$\mathbb{1}\{x^\top \alpha + \epsilon \geq p\}$	Log-concave noise and sparsity
Javanmard and Nazerzadeh (2019)	$\tilde{O}(s \log (d) \sqrt{T})$	i.i.d.	$\mathbb{1}\{x^\top \alpha + \epsilon \geq bp\}$	Log-concave noise and sparsity
Luo et al. (2024)	$\tilde{O}(dT^{\frac{2}{3}})$	Adver.	$\mathbb{1}\{x^\top \alpha + \epsilon \geq bp\}$	Log-concave noise with unknown distribution
Xu and Wang (2021)	$O(d \log T)$	Adver.	$\mathbb{1}\{x^\top \alpha + \epsilon \geq p\}$	Strictly log-concave noise
Ban and Keskin (2021)	$\tilde{O}(s \log (d) \sqrt{T})$	i.i.d.	$g(x^\top \alpha + x^\top \beta \cdot p) + \epsilon$	Sub-Gaussian noise and sparsity
Our paper	$\tilde{O}(d\sqrt{T})$	Adver.	$g(x^\top \alpha + x^\top \beta \cdot p) + \epsilon$	Sub-Gaussian noise

Notes. The notation $\tilde{O}(\cdot)$ omits the logarithmic factors, d is the dimension of the covariates and T is the horizon length. The column "Regret Bound" shows the regret upper bound, where s is the number of nonzero coordinates of unknown parameters when sparsity structure is assumed. The column "Covariates" describes the assumption on the generation of the covariates, where "Mart. Diff." stands for the covariates sequence has martingale difference and "Adver." means adversarial generations. In column "Demand Model," $\alpha, \beta \in \mathbb{R}^d$ and $b \in \mathbb{R}$, which are defaulted as fixed but unknown parameters without further comment in "Key assumptions," and ϵ is the unobserved error term. The column "Key assumptions" summarizes key assumptions for the corresponding paper.

Appendix B for more details). Let $\mathcal{X}$ denote the domain of x_t and $\theta^* := (\alpha^*, \beta^*)$ as the concatenated true parameter vector.

We make the following assumptions for the demand model and the pricing rule. The boundedness assumption is standard in the dynamic pricing literature, and it is a rather harmless one for practical applications. We note that the last one of the assumptions can be usually met by suitable normalization of the covariates.

Assumption 1 (Boundedness). *We assume*

a. *There exists* $\overline{\theta} > 0$ *such that* $\theta^* \in \Theta := \{\theta \in \mathbb{R}^{2d} : \|\theta\|_2 \leq \overline{\theta}\}$.

b. *The seller sets prices in a range* $[\underline{p}, \overline{p}]$ *for all* $t = 1, \ldots, T$.

c. *There exists a known* $\overline{D} > 0$ *such that* $g(x^\top \alpha^* + x^\top \beta^* \cdot p) \leq \overline{D}$ *for all* $x \in \mathcal{X}$ *and* $p \in [\underline{p}, \overline{p}]$.

d. *For all* $x \in \mathcal{X}$ *and* $p \in [\underline{p}, \overline{p}]$, $\|(x, px)\|_2 \leq 1$.

Assumption 2 (Properties of $g(\cdot)$). *The variable* $g(\cdot)$ *is strictly increasing and differentiable with a bounded derivative over its domain. Specifically, there exist constants* $\underline{g}, \overline{g} \in \mathbb{R}$ *such that* $0 < \underline{g} \leq g'(z) \leq \overline{g} < \infty$ *for all* $z = x^\top \alpha + x^\top \beta \cdot p$ *where* $x \in \mathcal{X}$, $\theta \in \Theta$, *and* $p \in [\underline{p}, \overline{p}]$.

Assumption 2 is a standard assumption for the link function of the generalized linear model (Filippi et al. 2010, Abeille and Lazaric 2017), and it ensures the learnability of the parameters (α, β).

Assumption 3 (Demand Shock). *Assume* $\{\epsilon_t, t = 1, 2 \ldots\}$ *forms a* $\overline{\sigma}^2$*-sub-Gaussian martingale difference sequence, that is,*

$$\mathbb{E}[\epsilon_t | \mathcal{H}_{t-1}] = 0 \textit{ and } \log(\mathbb{E}[e^{s\epsilon_t} | \mathcal{H}_{t-1}]) \leq \frac{\overline{\sigma}^2 s^2}{2}$$

for all $s \in \mathbb{R}$, *where* $\mathcal{H}_t := \sigma(p_1, \ldots, p_t, \epsilon_1, \ldots, \epsilon_t, x_1, \ldots, x_t, x_{t+1})$ *and* $\mathcal{H}_0 := \sigma(\emptyset, \Omega)$. *Moreover, we assume* $\overline{\sigma}^2$ *is known a priori.*

Assumption 3 on the error term covers common distributions such as normal distribution and distributions with bounded support. The sub-Gaussian parameter $\overline{\sigma}^2$ works as an upper bound for the true variance of the random variable. We note that the filtration definition includes the covariates at time $t + 1$. This small change allows the demand shock ϵ_t to be dependent on the covariates x_t at that time period and gives more modeling flexibility. We remark that, although such a martingale structure is assumed with respect to the demand shock, no assumption is imposed on the covariates x_t other than the boundedness in Assumption 1.

2.1. Performance Measure

For simplicity, we denote $a_t^* := x_t^\top \alpha^*$, $b_t^* := x_t^\top \beta^*$. We write the expected revenue function as

$$r(p; a, b) := p \cdot g(a + b \cdot p),$$

and the optimal expected revenue function as

$$r^*(a, b) := \max_{p \in [\underline{p}, \overline{p}]} p \cdot g(a + b \cdot p).$$

Throughout the paper, we assume this optimization problem can be efficiently solved.

Now, we define *regret* as the performance measure for the problem. Specifically,

$$\mathrm{Reg}_T^\pi(x_1, \ldots, x_T) := \sum_{t=1}^T r^*(a_t^*, b_t^*) - \mathbb{E}\left[\sum_{t=1}^T r_t^\pi\right],$$

where π denotes the online policy/algorithm in use, and the expectation is taken with respect to the demand

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shock ϵ_ts and the potential randomness from π. An admissible policy π maps the covariates $x_1, \ldots, x_t$, the past demand $D_1, \ldots, D_{t-1}$, and the past prices $p_1, \ldots, p_{t-1}$ to the price p_t^π offered at t. Accordingly, r_t^π denotes the revenue collected at time t under π. The benchmark oracle (the first summation in above) is defined based on the optimal revenue function $r^*(a_t^*, b_t^*)$. It assumes knowledge of $(a_t^*, b_t^*) = (x_t^\top \alpha^*, x_t^\top \beta^*)$ but does not observe the realization of the demand shock ϵ_t when setting the price. In defining the regret, we allow the covariates x_ts to be arbitrarily generated subject to Assumption 1. Therefore, no expectation is taken for x_ts in the regret definition, and we seek a worst-case regret upper bound over all possible x_ts. For the case when x_ts are i.i.d., the regret definition involves one more layer of expectation on x_ts. Thus our regret bound is stronger and can directly translate into a regret bound for the i.i.d. case.

3. Parameter Estimation and UCB-Based Pricing

In this section, we first introduce the maximum likelihood estimation for estimating the parameter θ^* based on the history data, which will be adopted in all pricing algorithms introduced later. The development of the approach is not new, and the presentation here aims to generate specific intuitions for dynamic pricing. Based on it, we present the prototypical UCB-based algorithm for the problem.

3.1. Maximum Quasi-Likelihood Estimator and Its Properties

From the demand model (1), we define the quasi-likelihood function (Wedderburn 1974) for the tth observation as follows:

$$l_t(\theta) := -\int_{D_t}^{g(z_t^\top \theta)} \frac{1}{h(u)}(u - D_t)\mathrm{d}u, \tag{2}$$

where $\theta = (\alpha, \beta)$ encapsulates the parameters, $z_t = (x_t, p_t x_t)$ is a column vector by concatenating the covariates, and $h(u) = g'(g^{-1}(u))$ for $u \in \mathbb{R}$. Because the distributions of the error terms (and thus the demands) are unknown, we cannot maximize the exact likelihood function and can only use the quasi-likelihood. To gain some intuition for $l_t(\cdot)$, we can treat it as the weighted mean of the gap between the observation D_t and the estimator $g(z_t^\top \theta)$: By noting $1/h(u) = (g^{-1})'(u)$, it can be interpreted as a weight that captures the first-order information (derivative) of θ revealed at point u. Then the function l_t as a first-order Taylor expansion proxy of $g^{-1}(\cdot)$ for the true likelihood function aims to find a θ that minimizes the weighted gap between $g(z_t^\top \theta)$ and D_t. To see this, when one maximizes l_t over θ, it achieves its maximum when $g(z_t^\top \theta) = D_t$ (if possible).

Based on l_t, we define the regularized quasi-likelihood estimator with parameter $\lambda > 0$ as

$$\hat{\theta}_t := \arg\max_{\theta \in \Theta} -\frac{\lambda \underline{g} \|\theta\|_2^2}{2} + \sum_{t'=1}^{t} l_{t'}(\theta), \tag{3}$$

where $\underline{g}$ denotes the lower bound of $g'(\cdot)$ as defined in Assumption 2. The estimator $\hat{\theta}_t$ will be used throughout the paper. The regularization term avoids the singularity caused by the arbitrariness of the covariates x_ts and also ensures a curvature for the likelihood function.

3.1.1. Estimation Error of $\hat{\theta}_t$. Define the (cumulative) sampled design matrix

$$M_t := \lambda I_{2d} + \sum_{t'=1}^{t} z_{t'} z_{t'}^\top,$$

where I_{2d} is a $2d$-dimensional identity matrix, and the M-norm of a vector z as $\|z\|_M := \sqrt{z^\top M z}$ for a positive definite matrix M. We use $\det M$ to denote the determinant of the matrix M. Then the proposition below bounds the estimation error of $\hat{\theta}_t$ according to the sampled design matrix M_t.

Proposition 1 (Filippi et al. 2010). *For any regularization parameter $\lambda > 0$, the following bound:*

$$\mathbb{P}\left(\exists t \in \{1, \ldots, T\} : \|\hat{\theta}_t - \theta^*\|_{M_t} \geq 2\sqrt{\lambda}\overline{\theta} + \frac{2\overline{\sigma}}{\underline{g}}\sqrt{2\log\left(\frac{1}{\delta}\right) + \log\left(\frac{\det M_t}{\lambda^{2d}}\right)}\right) \leq \delta$$

holds for any $\delta \in (0, 1)$.

The proposition is a key ingredient in removing the i.i.d. assumption on the covariates x_t. Specifically, if we impose an i.i.d. assumption on x_t and assume the minimum eigenvalue of its covariance matrix is bounded away from zero, then we can upper bound the estimation error of θ^* in the Euclidean norm. As we do not make such assumptions, the above proposition tells us that we can still obtain an estimation error bound by measuring the distance according to the sampled design matrix M_t. The following corollary makes the point even more precise by making the right-hand-side independent of M_t; we remark that this type of argument is commonly used in the linear bandits literature (Lattimore and Szepesvári 2020).

Corollary 1. *For all $\lambda > 0$,*

$$\mathbb{P}\left(\exists t \in \{1, \ldots, T\} : \|\hat{\theta}_t - \theta^*\|_{M_t} \geq 2\sqrt{\lambda}\overline{\theta} + \frac{2\overline{\sigma}}{\underline{g}}\sqrt{2\log(T) + 2d\log\left(\frac{2d\lambda + T}{2d\lambda}\right)}\right) \leq \frac{1}{T}.$$

3.2. UCB-Based Pricing

Algorithm 1 describes a UCB-based pricing algorithm that naturally arises from the estimator $\hat{\theta}_t$ in (3). At each

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time t, the algorithm first computes $\hat{\theta}_t$ and then constructs a confidence set based on the error bound in Corollary 1. Following the principle of optimism of UCB, the algorithm finds the most "optimistic" parameter within the confidence bound and sets the price pretending this parameter to be true. The algorithm design mostly mimics the GLM-UCB for generalized linear bandits (Filippi et al. 2010), and here it differs from GLM-UCB in a misalignment of rewards and observations. Specifically, the (realized) reward at period t collected from pulling the selected arm is just the observation at t in generalized linear bandits, and thus both the reward and observation are generalized linear functions. Here, the observations in the dynamic pricing setting contain the (realized) revenue and the demand at t, where the reward function is not a generalized linear function in action p_t anymore (although the demand is). Therefore, when estimating the unknown parameter θ^*, Algorithm 1 uses the observed demands by (3) instead of the observed rewards (revenues) as in Filippi et al. (2010). This small twist is natural but critical in achieving a tighter regret bound (see Online Appendix D.1 for more discussion).

Algorithm 1 (UCB Pricing)

Input: Regularization parameter λ.

for $t = 1, \ldots, T$ **do**

Compute the estimators $\hat{\theta}_{t-1}$ by (3) and its confidence set

$$\Theta_t := \left\{ \theta \in \Theta : \|\theta - \hat{\theta}_{t-1}\|_{M_{t-1}} \le 2\sqrt{\lambda}\overline{\theta} + \frac{2\overline{\sigma}}{\underline{g}} \sqrt{2\log(T) + 2d\log\left(\frac{2d\lambda + T}{2d\lambda}\right)} \right\}.$$

Observe covariates x_t and choose the UCB parameter that maximizes the expected revenue:

$$(\alpha_t, \beta_t) := \arg\max_{(\alpha,\beta)\in\Theta_t} r^*(x_t^\top \alpha, x_t^\top \beta). \tag{4}$$

Set the price by

$$p_t = \arg\max_{p\in[\underline{p},\overline{p}]} r(p; x_t^\top \alpha_t, x_t^\top \beta_t).$$

end for

Theorem 1 states the regret bound of Algorithm 1. For dimension d and horizon T, it meets the lower bound of the problem (Ban and Keskin 2021) up to logarithmic factors. Compared with the existing bounds (Qiang and Bayati 2016, Javanmard and Nazerzadeh 2019, Ban and Keskin 2021), our bound does not involve the term $\lambda_{\min}^{-1}$, where $\lambda_{\min}$ represents the minimum eigenvalue of the covariance matrix of x_t. For other parameters like $\overline{p}$ and $\overline{\theta}$, it is unclear whether their dependencies are optimal. We mention that they also appear in the existing regret bounds under the i.i.d. setting (Javanmard and Nazerzadeh 2019, Ban and Keskin 2021).

Theorem 1. *Under Assumptions 1, 2 and 3, with any sequence $\{x_t\}_{t=1,\ldots,T}$, if we choose the regularization parameter $\lambda = 1$, the regret of Algorithm 1 is upper bounded by*

$$4\bar{g}\bar{p}\bar{\gamma}\sqrt{Td\log\left(\frac{2d+T}{2d}\right)} + \overline{p}\overline{D} = \tilde{O}(d\sqrt{T}),$$

where $\overline{\gamma} = 2\overline{\theta} + \frac{2\overline{\sigma}}{\underline{g}}\sqrt{2\log T + 2d\log\left(\frac{2d+T}{2d}\right)}$ represents an upper bound for the confidence volume.

The proof largely mimics Filippi et al. (2010) except when leveraging the special structures of the pricing problem to deal with the fact that rewards (revenues) are not generalized linear functions in actions (prices) as discussed above. The analysis draws a connection between the dynamic pricing and the bandits problem and provides an alternative route for the existing analyses on the dynamic pricing with covariates problem (Qiang and Bayati 2016, Zhu and Zheng 2020, Ban and Keskin 2021, among others). As noted earlier, this new analysis relaxes the i.i.d. assumption and removes the dependency on the inverse minimum eigenvalue of the covariance matrix.

3.2.1. Tractability of the UCB Optimization Problem (4). We note that in general the optimization subroutine (4) can be nonconvex and hard to solve. In fact, computation efficiency is a common issue for UCB algorithms—the action selection step even for linear bandits can be NP-hard (Dani et al. 2008). One solution to compute the UCB optimization problem is through the Monte Carlo method: For each time t, the confidence set is an ellipsoid, so sampling from the confidence set can be done efficiently. Specifically, we randomly generate M samples from the uniform distribution over the confidence set and denote the samples as (α_m, β_m)s. Then the optimization subroutine (4) can be approximately solved by

$$(\alpha_t^{\mathrm{MC}}, \beta_t^{\mathrm{MC}}) = \arg\max_{m=1,\ldots,M} r^*(x_t^\top \alpha_m, x_t^\top \beta_m). \tag{5}$$

In the numerical experiments, we try out different values of M and examine the effect of sample size on the algorithm performance. With this Monte Carlo approach, the regret bound of Algorithm 1 will be revised by the following.

Theorem 2. *Under Assumptions 1, 2 and 3, with any sequence $\{x_t\}_{t=1,\ldots,T}$, if we choose the regularization parameter $\lambda = 1$, the regret of Algorithm 1 with a replacement of (α_t, β_t) by $(\alpha_t^{\mathrm{MC}}, \beta_t^{\mathrm{MC}})$ is upper bounded by*

$$4\bar{g}\bar{p}\bar{\gamma}\sqrt{Td\log\left(\frac{2d+T}{2d}\right)} + \overline{p}\overline{D} + \sum_{t=1}^{T} \mathbb{E}[r^*(x_t^\top \alpha_t, x_t^\top \beta_t) - r^*(x_t^\top \alpha_t^{\mathrm{MC}}, x_t^\top \beta_t^{\mathrm{MC}})],$$

where $\overline{\gamma} = 2\overline{\theta} + \frac{2\overline{\sigma}}{\underline{g}}\sqrt{2\log T + 2d\log\left(\frac{2d+T}{2d}\right)}$.

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The additional summation term in the regret bound captures the approximation error brought by the Monte Carlo method. Generally, a finer bound can be obtained for this approximation error with some additional structure of the underlying function r^* such as Lipschitzness. However, such a bound, or the Monte Carlo method itself, suffers from the curse of dimensionality (we defer more discussions to Online Appendix D.1).

Another source of computational suboptimality comes from solving the quasi-MLE problem (3). One cannot expect to solve the problem exactly in general, and an additional approximation error will also appear in the regret bound. We defer further discussions of this to Online Appendix D.3. To overcome the computational drawback of the UCB-based algorithm, we present a Thompson sampling-based algorithm in the next section.

4. Thompson Sampling-Based Pricing

In this section we devise a Thompson sampling-based method as a more efficient counterpoint to the UCB-based algorithm of the previous section. Ferreira et al. (2016) discussed the possibility of applying Thompson sampling for revenue management where the authors studied the covariate-free case and analyzed the constrained dynamic pricing/revenue management problem. Our result focuses on the handling of covariates and, more importantly, we analyze the problem under a frequentist rather than a Bayesian setting. A common belief is that such a frequentist regret bound will suffer from suboptimal regret that scales on the order of $d^{3/2}$ (Abeille and Lazaric 2017), with d being the number of unknown parameters compared with the linear-d dependency under the UCB-based algorithm. We show that such suboptimality is surprisingly avoidable for this dynamic pricing problem, which renders the Thompson sampling both computationally efficient and regret-optimal. Our finding is similar in spirit to that of Farias et al. (2022) on treatment effect estimation, although the analysis is different; both imply that the special structure such as a one-dimensional decision space is helpful in reducing the Thompson sampling's regret order under the frequentist regime.

4.1. Algorithm

Algorithm 2 presents the Thompson sampling-based pricing algorithm. The first step is the same as Algorithm 1, which is to solve the optimization problem (3) and obtain the estimator $\hat{\theta}_{t-1} = (\hat{\alpha}_{t-1}, \hat{\beta}_{t-1})$. The standard Thompson sampling algorithm (Abeille and Lazaric 2017) applies a randomized perturbation on the parameter $\hat{\theta}_{t-1}$ and obtains a perturbed estimation $\tilde{\theta}_{t-1} = \hat{\theta}_{t-1} + \eta_{t-1}$ with η_{t-1} being some Gaussian noise. Then the Thompson sampling algorithm will pretend the parameter $\tilde{\theta}_{t-1}$ as the true θ^* and set the price accordingly. The key design of our Thompson sampling algorithm is to first project the parameter onto the direction of x_t and obtain $(\hat{a}_t, \hat{b}_t) := (x_t^\top \hat{\alpha}_{t-1}, x_t^\top \hat{\beta}_{t-1})$. Next, instead of perturbing in the original parameter space, we perturb this projected parameter to obtain $(\tilde{a}_t, \tilde{b}_t)$. Recall that the matrix M_{t-1} measures the uncertainty of the current parameter estimation of $\hat{\theta}_{t-1} = (\hat{\alpha}_{t-1}, \hat{\beta}_{t-1})$. To measure the uncertainty of $(\hat{a}_t, \hat{b}_t)$, we should project the uncertainty matrix as well by

$$\tilde{M}_{t-1} := \left(\begin{pmatrix} x_t & \mathbf{0} \\ \mathbf{0} & x_t \end{pmatrix}^\top M_{t-1}^{-1} \begin{pmatrix} x_t & \mathbf{0} \\ \mathbf{0} & x_t \end{pmatrix} \right)^{-1}, \tag{6}$$

where $\mathbf{0}$ is the zero vector with dimension d. It is easy to verify that $\tilde{M}_{t-1}$ is well defined, that is, the matrix inside the parentheses is invertible. Essentially, $\tilde{M}_{t-1}$ guides the direction that we perform the perturbation in (7), and it shares the same intuition as the confidence set Θ_t in Algorithm 1.

Algorithm 2 (Thompson Sampling Pricing)

Input: Regularization parameter λ.
for $t = 1, \ldots, T$ **do**
Compute the estimator $\hat{\theta}_{t-1} = (\hat{\alpha}_{t-1}, \hat{\beta}_{t-1})$ by (3) and observe the covariates x_t.
Compute the estimator $(\hat{a}_t, \hat{b}_t) := (x_t^\top \hat{\alpha}_{t-1}, x_t^\top \hat{\beta}_{t-1})$.
Sample $\eta_t \sim \mathcal{N}(0, I_2)$ and compute the parameter

$$(\tilde{a}_t, \tilde{b}_t) := (\hat{a}_t, \hat{b}_t) + \tilde{\gamma}(\lambda) \tilde{M}_{t-1}^{-1/2} \eta_t. \tag{7}$$

where $\tilde{\gamma}(\lambda) = 2\sqrt{\lambda\theta} + \frac{2\overline{\sigma}}{\underline{g}} \sqrt{2 \log T + 2d \log\left(\frac{2d\lambda + T}{2d\lambda}\right)}$.
Set the price by

$$p_t = \arg\max_{p \in [\underline{p}, \overline{p}]} r(p; \tilde{a}_t, \tilde{b}_t),$$

and observe the demand D_t.
end for

4.2. Regret Analysis

To analyze the regret of Algorithm 2, we need a minor modification of Assumption 2.

Assumption 4 (Properties of $g(\cdot)$). *Let*

$$\tilde{\Theta} := \Bigg\{ \theta \in \mathbb{R}^{2d} : \|\theta - \tilde{\theta}\|_2 \le 4\overline{\theta} \sqrt{\log(4T^2)} + \frac{4\overline{\sigma}}{\underline{g}} \sqrt{\left(2 \log T + 2d \log\left(\frac{2d + T}{2d}\right)\right) \log(4T^2)} \text{ for some } \tilde{\theta} \in \Theta \Bigg\},$$

where Θ is as defined in Assumption 1. We assume $g(z)$ is strictly increasing and differentiable, and there exist constants $\underline{g}, \overline{g} \in \mathbb{R}$ such that $0 < \underline{g} \le g'(z) \le \overline{g} < \infty$ for all $z = x^\top \alpha + x^\top \beta \cdot p$ where $x \in \mathcal{X}$, $\theta = (\alpha, \beta) \in \tilde{\Theta}$, and $p \in [\underline{p}, \overline{p}]$.

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Assumption 4 is a stronger version of Assumption 2 on the properties of g in that it requires the same property of g to hold across a larger region $\tilde{\Theta}$ than Θ in Assumption 2. Specifically, the enlargement of Θ to $\tilde{\Theta}$ is to ensure the property of g covers the range of randomized sampled parameters in the Thompson sampling step.

The following lemma is adapted from lemma 3 of Abeille and Lazaric (2017). It implies that the sampled parameters $(\tilde{a}_t, \tilde{b}_t)$ are "optimistic" against the true parameter (a_t^*, b_t^*) with a certain probability for any convex function f. This probabilistic optimism is in parallel to the UCB design. The conditional part holds with high probability as in the previous UCB case.

Lemma 1. *Let $(a_t^*, b_t^*) = (x_t^\top \alpha^*, x_t^\top \beta^*)$. Then for the sampled parameter $(\tilde{a}_t, \tilde{b}_t)$ from (7) in Algorithm 2, the following inequality holds for any $t \geq 1$ and any function $f(a, b)$ that is convex in (a, b),*

$$\mathbb{P}(f(\tilde{a}_t, \tilde{b}_t) \geq f(a_t^*, b_t^*) \mid \mathcal{H}_{t-1}, (\alpha^*, \beta^*) \in \Theta_t) \geq \frac{1}{4\sqrt{e\pi}},$$

where

$$\Theta_t = \left\{ \theta \in \Theta : \|\theta - \hat{\theta}_{t-1}\|_{M_{t-1}} \leq 2\sqrt{\lambda}\overline{\theta} + \frac{2\overline{\sigma}}{\underline{g}} \sqrt{2\log(T) + 2d\log\left(\frac{2d\lambda + T}{2d\lambda}\right)} \right\},$$

is the confidence set defined in the UCB algorithm (Algorithm 1).

The above lemma is critical in our regret analysis and that of Abeille and Lazaric (2017). Yet there is a gap, as the revenue function is not convex in terms of the parameters (a, b). The following lemma bridges this gap. It basically says that the optimal revenue of the sampled parameters will be optimistic as long as the sampled parameters are optimistic for a linear function.

Lemma 2. *If $\tilde{a}_t + \tilde{b}_t \cdot p_t^* \geq a_t^* + b_t^* \cdot p_t^*$ where $p_t^* := \arg\max_{p \in [\underline{p}, \overline{p}]} r(p; a_t^*, b_t^*)$ is the optimal pricing at t, then $r^*(\tilde{a}_t, \tilde{b}_t) \geq \overline{r}^*(a_t^*, b_t^*)$ where recall $r^*(a, b) = \max_{p \in [\underline{p}, \overline{p}]} p \cdot g(a + b \cdot p)$.*

The lemma justifies the algorithm design of perturbing in the projected parameter space (a, b) rather than the original parameter space (α, β). There are two benefits to doing this. (i) It reduces the dimensionality of the perturbation and the regret bound; although we conduct parameter estimation in the original $2d$-dimensional space, the key sampling and analysis happen in this two-dimensional space, which is the reason for bound improvement. (ii) It relaxes the original requirement in Abeille and Lazaric (2017) that the reward function must be convex in the parameters, which limits the algorithm's application in the dynamic pricing context. Both benefits arise from the structure of the pricing and revenue function, and the proof of Lemma 2 is simple algebraic manipulation.

Theorem 3. *Under Assumptions 1, 3, and 4, with any sequence $\{x_t\}_{t=1,\ldots,T}$ and any $T \geq 6$, if we choose the regularization parameter $\lambda = 1$, the regret of Algorithm 2 can be bounded by*

$$36\overline{g}\overline{p}\overline{\gamma}\sqrt{2e\pi dT\log\left(\frac{2d+T}{2d}\right)\log(4T^2)} + 2\overline{D}\overline{p} = \tilde{O}(d\sqrt{T}),$$

where $\overline{\gamma} = 2\overline{\theta} + \frac{2\overline{\sigma}}{\underline{g}}\sqrt{2\log T + 2d\log\left(\frac{2d+T}{2d}\right)}$.

Theorem 3 gives the regret bound of Algorithm 2. The proof of the theorem mimics the analysis by Abeille and Lazaric (2017), and the key idea is to construct random parameters $(\tilde{a}_t', \tilde{b}_t')$ such that (i) the revenue $r^*(\tilde{a}_t', \tilde{b}_t')$ upper bounds $r^*(a_t^*, b_t^*)$ with high probability and (ii) the parameters $(\tilde{a}_t', \tilde{b}_t')$ connects to $(\tilde{a}_t, \tilde{b}_t)$ in a way that the single-step regret can be bounded by the volume of the confidence set (as the case of the UCB-based algorithm). Yet, in terms of the regret bound, we provide an analysis of the original Thompson sampling algorithm (Abeille and Lazaric 2017) for comparison in Online Appendix D.2, which results in a regret bound on the order of $d^{\frac{3}{2}}\sqrt{T}$. Furthermore, in comparison with Algorithm 1, Algorithm 2 achieves the same order of regret bound with a much smaller computation cost. In all, these results say that the Thompson sampling algorithm can be a computationally efficient substitute for the UCB-based algorithm without any compromise in the theoretical guarantee.

5. Pricing with Inventory Constraint

In the previous section, we consider the case where there is no inventory constraint. Here we proceed to a constrained setting. Specifically, consider the seller starts with an inventory of C. Let C_t be the remaining inventory at the beginning of time period t and $C_1 = C$. Accordingly, there is a minor twist in demand because of this inventory constraint. We define the realized sale/demand by

$$\tilde{D}_t := \min\{C_t, D_t\},$$

where D_t is the (true) demand in (1) and C_t follows the dynamic

$$C_t := C_1 - \sum_{t'=1}^{t-1} \tilde{D}_{t'}.$$

In other words, for the exact time period that the inventory is exhausted, the realized sale will be the demand D_t truncated by the available inventory.

The seller's objective is to maximize the expected revenue over the horizon

$$\mathbb{E}\left[\sum_{t=1}^{T} p_t^\pi \tilde{D}_t\right],$$

where the policy π specifies the price p_t^π.

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5.1. Model

For the case of unlimited inventory, whether the covariates are i.i.d. or non-i.i.d. do not significantly affect the best achievable regret bound. For the constrained case, however, it does make a difference. In this section, we consider a setting where the covariates are independent but nonidentical random vectors over time.

Assumption 5 (Independent Covariates). *The covariates $X_t \sim \mathcal{P}_t$ for $t = 1, \ldots T$, where $\mathcal{P}_t$s are unknown distributions. X_ts are independent of each other and also with the demand shocks ϵ_ts.*

Here we use the notation X_t to distinguish these random covariates from the deterministic yet arbitrarily chosen x_t in the previous sections. We remark that the assumption does not impose additional restrictions on the covariates than the arbitrarily chosen case. To recover the setting in Sections 3 and 4, we can set $\mathcal{P}_t$ to be point-mass distributions with $\mathbb{P}(X_t = x_t) = 1$. To the other extreme, if we have $\mathcal{P}_t = \mathcal{P}$, it recovers the setting of i.i.d. covariates. In this light, the aim of introducing $\mathcal{P}_t$s is to prepare for a finer characterization of the non-i.i.d. covariates but without any compromise on modeling flexibility.

As for the constrained case, we define the benchmark revenue as

$$\mathrm{OPT}(x_1, \ldots, x_T, C) := \max_{\pi \in \Pi_{\mathrm{hind}}} \mathbb{E}\left[\sum_{t=1}^{T} p_t^{\pi} \tilde{D}_t\right], \tag{8}$$

where Π_{hind} includes all pricing policies π that map

$$\{x_1, \ldots, x_T, D_1, \ldots, D_{t-1}, p_1, \ldots, p_{t-1}, \theta^*, C\}$$

to the price p_t^{π} offered at time t. Here the expectation is taken only with respect to the $\epsilon_t, t = 1, \ldots, T$ and the potential randomness of the pricing policy π. In defining the benchmark, the hindsight pricing policy is allowed to use the future covariates information and assumes the knowledge of the true model parameter θ^*. Accordingly, the regret of an online policy is defined by

$$\mathrm{Reg}_T^{\pi}(\mathcal{P}_1, \ldots, \mathcal{P}_T) := \mathbb{E}\left[\mathrm{OPT}(X_1, \ldots, X_T, C) - \sum_{t=1}^{T} p_t^{\pi} \tilde{D}_t\right],$$

where the expectation is taken with respect to the demand shock ϵ_ts, covariates X_t, and potential randomness in π. The online policy $\pi \in \Pi_{\mathrm{on}}$ does not observe future covariates information $\{x_{t'}\}_{t'=t+1}^{T}$ or know the true parameter θ^*.

We introduce the following measure to quantify the variation of the distributions over time:

$$\mathcal{W}_T(\mathcal{P}_1, \ldots, \mathcal{P}_T) := \sum_{t=1}^{T} \mathcal{W}(\mathcal{P}_t, \bar{\mathcal{P}}_T),$$

where $\mathcal{W}(\cdot, \cdot)$ is the Wasserstein distance function to measure the distance between two distributions (Kantorovich 1960) with the Euclidean distance as the underlying metric and $\bar{\mathcal{P}}_T := \frac{\sum_{t=1}^{T} \mathcal{P}_t}{T}$, that is, the uniform mixture of $\mathcal{P}_t$s. When $\mathcal{P}_t$s are identical for $t = 1, \ldots, T$, then $\mathcal{W}_T(\mathcal{P}_1, \ldots, \mathcal{P}_T) = 0$. This measure has been used to quantify the intensity of non-i.i.d.-ness or nonstationarity of online optimization algorithms under a full information environment (Jiang et al. 2020) and under a partial information environment (Liu et al. 2022).

Theorem 4 (Regret Lower Bound). *The following lower bound holds for any policy $\pi \in \Pi_{\mathrm{on}}$:*

$$\mathrm{Reg}_T^{\pi}(\mathcal{P}_1, \ldots, \mathcal{P}_T) = \Omega(\max\{\sqrt{T}, \mathcal{W}_T(\mathcal{P}_1, \ldots, \mathcal{P}_T)\}).$$

Theorem 4 gives a lower bound of the regret that is $\Omega(\max\{\sqrt{T}, \mathcal{W}_T(\mathcal{P}_1, \ldots, \mathcal{P}_T)\})$. The $\Omega(\sqrt{T})$ part is directly implied from the existing lower bound for the previous unconstrained dynamic pricing problems (Broder and Rusmevichientong 2012, Keskin and Zeevi 2014), which are special cases of the constrained ones where the initial inventory $C \to \infty$. The $\Omega(\mathcal{W}_T(\mathcal{P}_1, \ldots, \mathcal{P}_T))$ term is exactly the variation measure above. In the unconstrained case, we note that the regret is affected by the covariates x_ts through the bound on its norm. In contrast, for this constrained case, the variation in the covariates does affect the regret lower bound through the term $\Omega(\mathcal{W}_T(\mathcal{P}_1, \ldots, \mathcal{P}_T))$. The intuition for this contrast is that, for the unconstrained setting, the variation in distribution only affects the learning procedure, that is, the estimation of the true model parameter. For the constrained setting, the variation results in an additional challenge in balancing the resource consumption: The seller does not know whether the future periods will be more profitable or less and hence whether to reserve more inventory or less for the future periods.

5.2. Algorithm

To generate some intuition for the algorithm, we first write out the deterministic optimization problem for the pricing problem,

$$\begin{aligned} \max_{p_1, \ldots, p_T} \quad & \sum_{t=1}^{T} \mathbb{E}[p_t \cdot g(A_t + B_t \cdot p_t)] \\ \text{s.t.} \quad & \sum_{t=1}^{T} \mathbb{E}[g(A_t + B_t \cdot p_t)] \le cT, \end{aligned} \tag{9}$$

where $A_t = X_t^{\top} \alpha^*$ and $B_t = X_t^{\top} \beta^*$. Here $c = C/T$ denotes the average available inventory per time period. The optimization problem replaces the random demand with the corresponding expectation and only requires the constraint to be satisfied on expectation. It is easy to verify that the optimal value of this deterministic program upper bounds the hindsight benchmark $\mathbb{E}[\mathrm{OPT}(X_1, \ldots, X_T, C)]$. This upper bound relationship makes the deterministic program a nice starting point for algorithm development and theoretical analysis in

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the literature (Gallego and Van Ryzin 1997, Talluri et al. 2004).

The Lagrangian of (9) is

$$L(\mu, p_{1:T}) := c\mu T + \sum_{t=1}^{T} \mathbb{E}[(p_t - \mu) \cdot g(A_t + B_t \cdot p_t)],$$

where the primal decision variables are $p_1, \ldots, p_T$, and the dual decision variable is μ. Such a formulation inspires the development of dual-based algorithms (Agrawal and Devanur 2016, Li and Ye 2022).

First, let μ^* be the optimal dual solution, and then the optimal primal price should accordingly be determined by

$$p_t^* = \arg\max_{p_t \in [\mu^*, \overline{p}]} (p_t - \mu^*) \cdot g(A_t + B_t \cdot p_t). \tag{10}$$

Intuitively, the additional term μ^* reflects the cost arising from the limited inventory and can be interpreted as the shadow price. Also, we remark that the price in (9) should be a function of the covariates X_t; we omit this dependency in the notation for simplicity without hurting the generality.

Second, if we see the primal price variables as fixed and focus on the dual variable, the Lagrangian becomes

$$L(\mu, p_{1:T}) = \mu \cdot \sum_{t=1}^{T} \mathbb{E}[c - g(A_t + B_t \cdot p_t)] + \sum_{t=1}^{T} \mathbb{E}[p_t \cdot g(A_t + B_t \cdot p_t)], \tag{11}$$

where only the first summation involves the dual variable.

Algorithm 3 implements the Thompson sampling algorithm for this constrained setting. The parameter learning part is the same as Algorithm 2. The price optimization step (12) optimizes (10) using μ_t as an estimate of μ^*. The update of the dual price μ_t in (13) essentially performs a projected online gradient descent with respect to (11), and the step size is chosen to be $\frac{\overline{p}}{\sqrt{(\overline{D}^2+\overline{\sigma}^2)T}}$.

Algorithm 3 (Thompson Sampling with Inventory Constraint)

Input: Regularization parameter λ, initial dual variable $\mu_1 = 0$.

for $t = 1, \ldots, T$ **do**

Compute the estimator $\hat{\theta}_{t-1} = (\hat{\alpha}_{t-1}, \hat{\beta}_{t-1})$ by (3) and observe the covariates $X_t = x_t$.

Compute the estimator $(\hat{a}_t, \hat{b}_t) := (x_t^\top \hat{\alpha}_{t-1}, x_t^\top \hat{\beta}_{t-1})$.

Sample $\eta_t \sim \mathcal{N}(0, I_2)$ and compute the parameter

$$(\tilde{a}_t, \tilde{b}_t) = (\hat{a}_t, \hat{b}_t) + \left(2\sqrt{\lambda}\overline{\theta} + \frac{2\overline{\sigma}}{\underline{g}}\sqrt{2\log T + 2d\log\left(\frac{2d\lambda + T}{2d\lambda}\right)}\right)\tilde{M}_{t-1}^{-1/2}\eta_t.$$

Set the price by

$$p_t = \arg\max_{p \in [\mu_t, \overline{p}]} (p - \mu_t) \cdot g(\tilde{a}_t + \tilde{b}_t \cdot p), \tag{12}$$

and observe the demand D_t.

If the remaining inventory cannot fulfill the demand, that is, $D_t \geq C_t$, then the algorithm stops; otherwise, update the dual variable:

$$\mu_{t+1} = \mathrm{Proj}_{[0,\overline{p}]}\left(\mu_t + \frac{\overline{p}}{\sqrt{(\overline{D}^2 + \overline{\sigma}^2)T}} \cdot (D_t - c)\right) \tag{13}$$

and the remaining inventory

$$C_{t+1} = C_t - D_t.$$

end for

5.3. Regret Analysis

For regret analysis, we introduce a few additional assumptions to ensure feasibility and tractability.

Assumption 6 (Feasibility and Tractability). *We assume*

a. *The average inventory per time period* $c = C/T \in [0, \overline{D}]$.

b. *For all* $x \in \mathcal{X}$, *we have* $x^\top \beta^* < 0$ *and* $g(x^\top \alpha^* + x^\top \beta^* \cdot \overline{p}) < c$.

c. *The function* $d \cdot g^{-1}(d)$ *is a convex function for* $d \in [0, \overline{D}]$.

Part (a) and Part (b) ensure that the constrained problem is well defined. Part (c) ensures the single-period pricing problem can be efficiently solved and is a standard assumption in the revenue management literature (Gallego and Van Ryzin 1997, Chen and Gallego 2022, Chen and Shi 2023) and holds for many $g(\cdot)$s, including $g(x) = x$, $g(x) = x^2$ and $g(x) = \exp(x)$.

Theorem 5 (Regret Upper Bound). *Under Assumptions 1, 3, 4, 5, and 6 with* $T \geq 6$, *if we choose the regularization parameter* $\lambda = 1$, *the regret of Algorithm 3 can be bounded by*

$$\tilde{O}(d\sqrt{T} + \mathcal{W}_T(\mathcal{P}_1, \ldots, \mathcal{P}_T)).$$

To interpret the regret bound, we first make a comparison against Theorem 4. The variation measure $\mathcal{W}_T$ term appears in both the lower and upper bounds, and thus it provides a tight characterization of how the change of distribution affects the algorithm performance. Next, when the distributions $\mathcal{P}_t$s are identical to each other, the $\mathcal{W}_T$ term disappears and the result recovers the existing bounds for the i.i.d. setting (Keskin and Zeevi 2014, Li and Zheng 2024).

We note the regret bound under the partial-information setting of the pricing problem (i) matches the one for full-information online constrained stochastic optimization (Jiang et al. 2020) and (ii) is strictly better than the one for bandits with knapsacks (BwK) (Liu et al. 2022), which is partial-information setting as well. This is because, in our partial-information pricing

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problem, the underlying demand model does not change over time, that is, (α^*, β^*) remains the same over time; only the covariates distribution $\mathcal{P}_t$ changes over time. If the demand model was to change over time, the regret bound above will become aligned with the BwK regret bound of Liu et al. (2022). Finally, in comparison with the previous unconstrained setting, the results in this section show that the variation in the covariates distribution substantially affects the algorithm performance under the constrained setting.

6. Numerical Experiments

We consider the linear demand model $D_t = x_t^\top \alpha^* + x_t^\top \beta^* \cdot p_t + \epsilon_t$ with two groups of numerical experiments:

a. Without inventory constraint: we implement and compare the performance of Algorithm 1 (with different numbers of Monte Carlo samples), Algorithm 2, and benchmark algorithms.

b. With inventory constraint: we implement Algorithm 3 and compare its performance against benchmark algorithms. We also examine the effect of different $\mathcal{W}_T$s on the regret.

All the numeric results (such as the curves in the plot) are reported based on the average performance over 100 simulation trials. More experiment details are deferred to Online Appendix E.

6.1. Without Inventory Constraint

6.1.1. Environment Setup. We set the horizon $T = 1{,}500$. The whole horizon is split into two phases with equal length. In the first phase, the first half of the covariates (dimension 1 to $d/2$) are i.i.d. generated over time, whereas the second half (dimension $d/2+1$ to d) are all zero; in the second phase, the first half of the covariates (dimension 1 to $d/2$) are all zero, whereas the second half (dimension $d/2+1$ to d) are i.i.d. generated over time. Moreover, we set the allowable price range as $[0.1, 5]$. For each simulation trial, the true parameter α^* is generated by a uniform distribution over $\frac{1}{\sqrt{d}}[1,2]^d$, the true parameter β^* is generated by a uniform distribution over $-\frac{1}{\sqrt{d}}[0,1]^d$, and the covariates (if nonzero) are always generated i.i.d. from a uniform distribution over $\frac{1}{\sqrt{d}}[0,1]$. The normalizing factor $\frac{1}{\sqrt{d}}$ ensures the demand always stays on the same magnitude for different d.

6.1.2. Results. Figure 1, (a) and (b), compares the cumulative regrets of Algorithm 1 (UCB) with different Monte Carlo samples, and Algorithm 2 (TS), with two benchmarks under $d = 6$ and $d = 12$, respectively. The first benchmark algorithm is Algorithm 5 (see Appendix D.2 for details), the original Thompson sampling (TS_Ori), which does the random perturbation in the original parameter space instead of the projected space. For the second benchmark, we adapt the covariate-free constrained iterated least square (CILS) algorithm (Keskin and Zeevi 2014) designed for the i.i.d. covariates setting. We refer to Online Appendix E for more details about CILS. We notice that the CILS algorithm performs well for the first phase but fails to learn the parameter during the second phase. UCB performs stably well under different numbers of Monte Carlo samples. At the beginning of the second phase, the curves of UCB and TS grow quickly but then they all flatten. In addition, TS outperforms the original Thompson sampling TS_Ori in both figures, which is consistent with the improved factor of $\sqrt{d}$ in our theoretical regret bound.

6.2. With Inventory Constraint

6.2.1. Environment Setup. We also consider splitting the horizon T into two phases with equal length. The covariates follow three patterns with different variation measure $\mathcal{W}_T$s:

- Large: the covariates are generated i.i.d. from a uniform distribution over $\frac{0.1}{\sqrt{d}}[0,1]$ in the first phase and $\frac{5}{\sqrt{d}}[0,1]$ in the second phase, which gives the largest $\mathcal{W}_T$ among the three patterns.
- Small: the covariates are generated i.i.d. from a uniform distribution over $\frac{1}{\sqrt{d}}[0,1]$ in the first phase and

Figure 1. (Color online) Experiment a: Performances of Algorithms Without Inventory Constraint

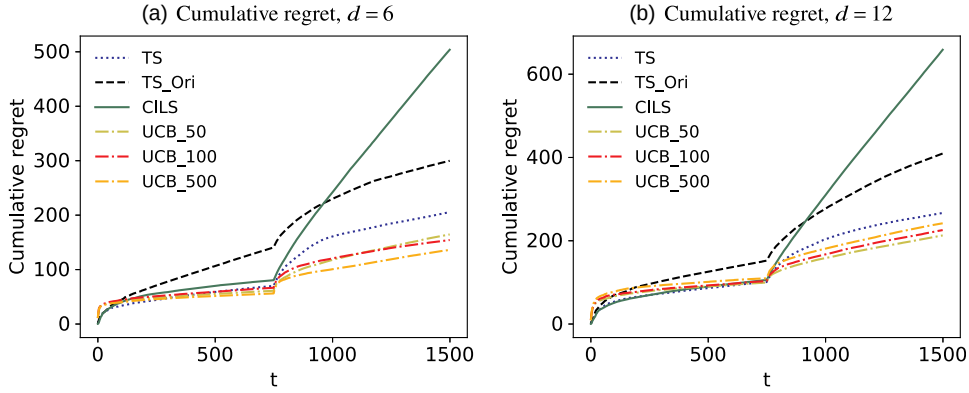

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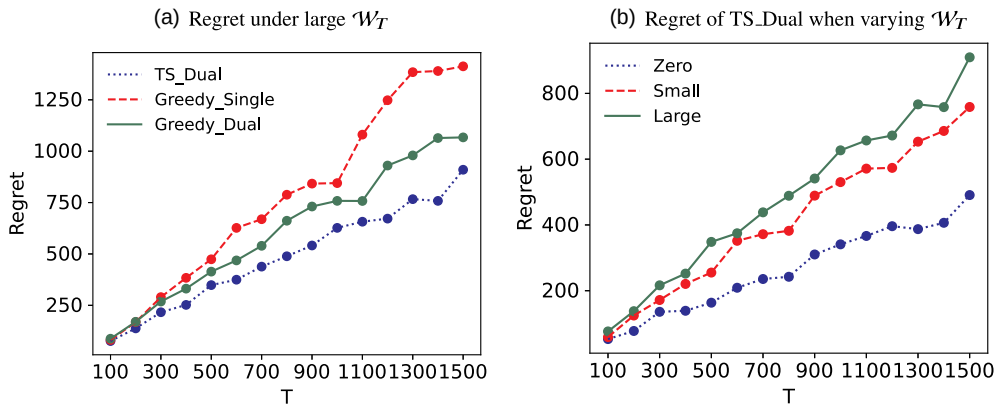
Figure 2. (Color online) Experiment b: Performances of Algorithms with Inventory Constraint



$\frac{5}{\sqrt{d}}[0,1]$ in the second phase, where $\mathcal{W}_T$ is smaller than the case of Large.

- Zero: the covariates are generated i.i.d. from a uniform distribution over $\frac{5}{\sqrt{d}}[0,1]$ in both phases, where the variation measure $\mathcal{W}_T = 0$.

We set the averaged inventory over periods as $c = C/T = 0.5$. As a reference point, the averaged optimal demands for the above three patterns when there is no inventory constraint are approximately 1, 1.2, and 1.9, respectively. We set the dimension of covariates $d = 6$. The generations of θ^* and the range of pricing remain the same as before.

6.2.2. Results. Figure 2(a) reports the performance of Algorithm 3 (TS_Dual) over $T = 100, 300, \ldots 1{,}500$ with two other benchmarks under the Large pattern. The benchmark algorithm Greedy_Single solves the deterministic single-period pricing problem under inventory c with using $\hat{\theta}_t$ as θ^* and algorithm Greedy_Dual is the same as TS_Dual except replacing $(\tilde{a}_t, \tilde{b}_t)$ by $(\hat{a}_t, \hat{b}_t)$, that is, without the Thompson sampling step. More details of these two algorithms can be found in Online Appendix E. Figure 2(a) shows TS_Dual outperforms two benchmark algorithms consistently for different Ts. Furthermore, its regret curve does not scale with $O(\sqrt{T})$. This verifies the results in Theorem 5, where the regret upper bound includes an additional term $\mathcal{W}_T$, which also depends on T. We further examine the influence of $\mathcal{W}_T$ on the regret. We plot the regrets of TS_Dual over $T = 100, 300, \ldots 1{,}500$ under different patterns in Figure 2(b). We observe that the regrets are consistently increasing in $\mathcal{W}_T$ for different Ts, which numerically coincides the results in Theorem 5.

7. Conclusion

In this paper, we consider the dynamic pricing problem under a generalized linear demand model where the demand is dependent on the price and arbitrary covariates. We propose UCB-based and Thompson sampling-based pricing algorithms inspired by bandits literature, which can both achieve the optimal order of regret by leveraging the special structure of the pricing problem. We further consider the pricing problem with inventory constraint and show the Thompson sampling algorithm can be extended to this case.

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